

IRIS-HEP Summer Fellow Project Proposal

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Project Title: Building an Acknowledgement Plugin for the Alarms and Alerts System

Duration: June 2026 - August 2026 (12 weeks)

Project Description

perfSONAR is a widely used monitoring framework for research and education networks that support global scientific collaboration. The Alarms and Alerts system processes millions of daily network tests and generates alarms that notify site administrators and scientists about network issues across different communities.

Currently, when an issue is detected, the system sends email notifications to all subscribed users. If the issue persists, repeated emails continue to be sent every time the analysis runs - even if someone is already actively working on the problem.

This project proposes building an acknowledgement plugin that allows administrators to:

- Acknowledge an alarm
- Add a comment
- Suppress further email notifications for all subscribers for a specified time period

If time allows this project will also explore adding new alarm types to expand the system's diagnostic capabilities.

Goals

The main goal is to design and implement an acknowledgement feature that integrates with the existing alarm and notification infrastructure.

Key design questions include:

- How users will interact with the acknowledgement system (UI, workflow)
- How acknowledgements will be linked to alarms in the database schema
- What metadata should be stored (user identity, comments, timestamps, duration)
- How the notification system should check acknowledgement status before sending emails

Proposed Timeline

Weeks 1-3: Understanding the System & Design

- Study the existing alerting infrastructure
- Understand:
 - How alarm data is stored and queried
 - How users subscribe to alarms and receive notifications
- Set up the local development environment
- Design the acknowledgement feature:
 - Database schema
 - User interface (acknowledge, comment, time period)
 - Notification suppression logic

Weeks 4-6: Main Implementation

- Implement backend logic:
 - Data model for acknowledgements
 - API endpoints for creating and managing acknowledgements
- Build frontend interface:
 - Acknowledge alarm button/action
 - Comment field
 - Suppression time configuration
- Integrate with notification system:
 - Suppress emails for acknowledged alarms
 - Resume notifications after expiration
- Testing:
 - Trigger alarms
 - Validate suppression behavior
 - Verify that emails start sending alerts again after the set time period ends

Week 7: Stabilization

- Fix bugs and handle edge cases
- Present mid-project progress and findings

Weeks 8-9: Deployment Preparation

- Improve feature based on feedback
- Add visibility features:
 - Show acknowledgment status
 - Display who acknowledged the alarm
 - Show expiration time
- Prepare integration and deployment into production infrastructure

Weeks 10-11: New Alarm Types

- Explore additional alarm types
- Develop and test new alarm scripts
- Deploy and validate new monitoring signals

Week 12: Finalization

- Evaluate system performance in real-time conditions
- Refine features as needed
- Document the full project
- Deliver final presentation to mentors and fellows

References

- <https://aaas.atlas-ml.org/>
- <https://github.com/ATLAS-Analytics/AlarmAlertServiceFrontend/tree/main>
- <https://ps-dash.uc.ssl-hep.org/>
- <https://atlas-kibana.mwt2.org/s/networking/app/discover>
- <https://www.perfsonar.net/>